



ABRAR MUHTASIM PATHAN

Urban and Rural Planning Student and Researcher



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PROFILE INFO

Graduated from Urban and Rural Planning Discipline of Khulna University specialized in GIS and Remote Sensing. Previously worked on several spatial analysis and social science projects including multi-hazard mapping, susceptibility mapping by performing analysis in tools like ArcGIS Pro, Google Earth Engine, SPSS and Python.

PROJECTS AND WORK EXPERIENCE

- Internship at Bangladesh Space Research and Remote Sensing Organization (SPARRSO) December 2025**
 - Boundary delineation of Kaptai Lake using Sentinel-1 and Sentinel-2
 - Report writing on SPARRSO's previous and existing projects
 - Receiving satellite data and processing in ArcGIS
 - Presentation on Kaptai Lake delineation project
- Vulnerability assessment in coastal region of Bangladesh November 2024– July 2025**

The objective of the work was to detect vulnerable households and its extent from the lens of intersectionality. In this regard primary data was collected from 200 households. Coastal union from Satkhira district was the research area for the study.

Contribution Statement: Conceptualization | Data collection | Formal Statistical Analysis | Writing

Applied tools: Excel | SPSS
- Landslide Susceptibility Mapping (Q1) April 2025 – Ongoing**

Predicting landslide susceptible zones in Chittagong hilly region using machine learning techniques. The work is based on secondary data. Previously landslide events was taken to train ML model and predict future outcome.

Contribution Statement: Data collection | Formal Analysis

Applied tools: GIS | Earth Engine | Python | ML
- Flood Susceptibility Mapping using SAR data (Undergrad Thesis) May 2025 – Ongoing**

Susceptibility mapping of critical river basin in northern region of Bangladesh during the time of flood using SAR data and machine learning based models.

Contribution Statement: Conceptualization | Data collection | Formal Analysis | Writing

Applied tools: GIS | Earth Engine | Python | ML
- Earthquake based Multi Hazard mapping in Bangladesh June 2025 – Ongoing**

Mapping of earthquake hazard zone using MCDA and AHP method to see the extent of earthquake affected areas. The work is based on secondary data.

Contribution Statement: Data collection | Formal Analysis | Writing

Applied tools: GIS | Earth Engine

PROJECTS AND WORK EXPERIENCE

- Urban Hit Island detection in major cities of Europe

June 2025 – September 2025

The objective of the work is to determine the impact of urbanization and greeneries in generating and mitigation of urban hit island.

Contribution Statement: Team lead | Data collection | Formal Statistical Analysis | Writing

Applied tools: Google Earth Engine | SPSS | GIS
- Deep Learning based object detection in traffic management of Bangladesh

January 2024 – September 2024

Detection of different classes of vehicle in urban traffic management context

Applied tools: Python | Deep learning

Contribution Statement: Data collection | Graphical visualization | Writing
- Master plan of Pankhali union in Dacope subdistrict (Academic Work)

September 2024 – November 2024

The work was based on a union of Khulna district. The master plan was a project of my undergraduate academic session where I was in the lead of the team and secured highest grade(A) in the project.

Contribution Statement: Team lead | Project design | Data collection | Formal Statistical Analysis | Writing

Applied tools: Excel | SPSS | GIS | KOBO
- Travel behavior report on Khulna City Corporation (Academic Work)

September 2024 – November 2024

The objective of the work was to determine travel pattern of citizens living in Khulna City Corporation ward boundary. In this purpose total of 800 household survey have been conducted to collect the travel pattern data of the household members. Furthermore, statistical analysis have been conducted to determine the extent of different variables.

Contribution Statement: Data collection | Formal Statistical Analysis | Writing

Applied tools: Excel | SPSS | GIS
- Designing whole thesis project with time-period on Micro-Soft project software (Academic Work)

August 2025

The work was based on the thesis work and activities and designing it through Microsoft project to visualize the workflow and keeping the track in time.

Applied tools: Micro-Soft Project
- Cyclone based Multi Hazard Mapping in Bangladesh

September 2024 – November 2024

Mapping of cyclone hazard zone using MCDA(Multi-Criteria Decision Analysis) and AHP(Analytical Hierarchy Process) method to see the extent of cyclone affected areas in the Chittagong City Corporation. Work is based on secondary data.

Contribution Statement: Conceptualization | Data collection | Formal Analysis | Writing

Applied tools: GIS | Earth Engine | Python

EXTRA CURRICULAR ACTIVITIES

- Sergeant at arms in Rotaract Club of Khulna University

February 2022– June 2024

 - Active contribution as a part of management committee in more than 30+ sessions
 - Arranger of Job Fair of two sessions.
- Member of Student Square (A non-profit societal fostering organization)

September 2024– Ongoing

 - Speaker and as a management contributor in school counselling sessions
 - Actively designed presentations regarding sessions for student nurturing
 - Providing innovative solutions on expanding organization’s output visualization
 - Performing statistical analysis to generate model to visualize organization output

EXTRA CURRICULAR ACTIVITIES

- **Co-founder and treasurer of PLANNEX (University Club)** May 2025– Ongoing
 - Successfully arranged sessions on career talks with participants of more than 160 people.
 - Managing the financial transactions and account of the club

ACHIEVEMENT AND AWARDS

- **Champion of REAL-ESTATE idea generation presentation competition** February 2025
 - The competition was based generating innovative housing solution ideas and presenting them in-front of national level judges

Contribution Statement: Team lead | Presentation design | Data collection | Presentation delivery | Writing

Applied tools: Excel | Power-point
- **Academic excellence award** Session 2023-2024
 - Recognition from department for getting CGPA more than 3.5
- **9th position in Intra university quiz competition** April 2022
 - Actively participated in quiz competition as a participant and secured in top 10 among the whole university

CERTIFICATION

- **Participation as an organizer and participant in skill development session by Rotaract Club of Khulna University** June 2023
 - The session was taken by Sadman Sadik(IBA graduated influencer).
- **Participation in strengthening urban climate resilience of Khulna towards achieving SDGs by 2030** June 2023
 - The session was organized by GIZ Bangladesh and BYEI(Bangladesh Youth Environmental Initiative)
- **Participation in GIS competition by KUET** February 2025
 - In Plannation 3.0, I have participated as a participant in GIS competition where the judges were the professor from KUET, BUET, RUET and KU.
- **Completion of project essential course by disaster** February 2025
 - In Plannation 3.0, I have participated as a participant in GIS competition where the judges were the professor from KUET, BUET, RUET and KU.

SOFTWARE SKILLS

Software and Tools	Work ability
ArcGIS	GIS Analysis (site suitability, overlay, buffer, proximity, etc.), Geoprocessing ,Spatial Analysis & Modeling, Overlay & Weighted Overlay Analysis, Hotspot & Coldspot Analysis, Topology and Network Analysis, Land Use/Land Cover Analysis, Multi-Criteria Decision Analysis, Site Suitability Assessment, Georeferencing, Map Digitization from Raster to Vector format., Google Earth Mapping
Google Earth Engine	Image Classification (Unsupervised, Supervised, Object-Based, etc., NDVI,SAVI,ENVI,DEM and other indices
AutoCAD	2D layout design
Sketch Up	3D layout design, rendering
PTV Vissim	3D Traffic modelling
SPSS	Statistical inferential and descriptive analysis and visualization
Python	Python based data analysis and visualization in different formats with more precision

EDUCATION

2022 - 2025 | Khulna University
Bachelor of Urban and Rural Planning
CG: 3.52 / 4.00 (Got general scholarship in all semesters)

LANGUAGES

- English (Fluent | IELTS – 7.5)
- Bangla (Native)
- Hindi (Communication)

REFERENCE

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